



a clerk's position in the accounts section of the Madras Port Trust. The chief accountant of the port trust, S. Narayana Rao, was a mathematician. Both he and Sir Francis Spring, the chairman of the port trust, took a keen interest in Ramanujan's mathematical talents.

The mathematicians in Madras persuaded Ramanujan to contact experts in England to evaluate his work, but given his lack of formal education, he was not taken seriously and his letters went largely unacknowledged. In 1913, however, Ramanujan sent an intriguing letter to G.H. Hardy, an eminent mathematics professor at Trinity College, Cambridge. In the letter, he posited about 120 mathematical theorems without showing the details of how he had acquired the results. "I had never seen anything like them before," Hardy wrote later. "A single look at them was enough to show that they could be written by a mathematician of the highest class. They must be true because, if they were not true, no one would have the imagination to invent them". Hardy was so impressed that he suggested that Ramanujan come to Cambridge for further study. Hardy and his colleague, J.E. Littlewood, made special efforts to admit Ramanujan to Cambridge University, despite his lack of a degree in mathematics.

Initially, there was opposition to Ramanujan's going abroad given his strict religious background. According to some, his mother had a dream in which the Goddess Namagiri commanded her not to stand in the way of her son's goals. Thereafter, with his family's blessings, Ramanujan arrived at Cambridge in 1914. His research flourished and he published many exciting new results on topics such

Q: What is the *Ramanujan Journal*?

A: An international publication about his work, launched in 1997

Q: How many identities and equations did he compile?

A: 3,900

Q: When and how did he become famous?

A: In 1913, he wrote a 10-page letter to mathematician G.H. Hardy

Q: When did he master advanced trigonometry?

A: At the age of 13

Q: In which fields have his discoveries become applicable?

A: Crystallography and string theory

as the number theory, infinite series and indefinite integrals. One of the most spectacular results in mathematics is the Hardy-Ramanujan formula derived in 1917 for the number of partitions of an integer. A striking characteristic of Ramanujan's work is the mysterious mix of symbols and formulas. He believed that the Goddess Namagiri appeared in his dreams to guide and inspire his work.

Ramanujan was awarded a degree from Cambridge in 1916 and was made a Fellow of the Royal Society (FRS) in 1919. Perhaps due to the intense pressure of work and lack of a proper diet, he contracted tuberculosis in England and was admitted to a nursing home. Hardy visited him there and remarked, "I thought the number of my taxicab was 1729, it seemed to me a rather dull number." Ramanujan replied:

"No Hardy! It is a very interesting number. It is the smallest number expressible as the sum of two cubes in two different ways". Ramanujan returned to India in 1919 and died in Kumbhakonam the next year.

He was richly lauded for his achievements. His name hit the headlines of newspapers when he was awarded a degree by Cambridge. A postage stamp commemorating his 75th birth anniversary was issued in 1962. Various mathematicians have devoted their lives trying to decipher Ramanujan's work from his notebooks. Conferences are held worldwide devoted to ideas inspired by him. He has been compared to some giants in the field for the beauty and originality of his ideas.

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